

Proposed New Division- KIPRE

Contents

2.7.3. Data Science and Analytics Division.....	1
a) Rationale	1
b) Functions.....	1
c) Organizational Structure	2
1. Biostatistics and Statistical Modelling Section.....	2
a) Rationale	2
b) Functions.....	3
2. Bioinformatics and Computational Analytics Section.....	3
a) Rationale	3
b) Functions.....	3
3. Data Analytics and Decision Support Section	4
a) Rationale	4
b) Functions.....	4
4. Staffing	5

2.7.3. Data Science and Analytics Division

a) Rationale

The Data Science and Analytics Division draws its mandate from Section 5 of Legal Notice No. 273 of 2017 (ST&I Act, No. 28 of 2013), which requires the Institute to undertake research and associated capacity support.

The Division will provide a strategic, cross-cutting scientific and analytical capability applying advanced statistical, computational and data science methods to genomic, clinical, epidemiological and laboratory data across research, surveillance and innovation programs.

It will strengthen institutional capacity in biostatistics, statistical and computational modelling, bioinformatics, Artificial Intelligence (AI), Machine Learning (ML) and advanced data analytics, while supporting capacity building through training, mentorship and knowledge transfer. The Division will provide the technical leadership and specialized expertise required to advance data-driven research, innovation, surveillance and evidence-informed decision-making.

b) Functions

The Division for Data Science and Analytics is charged with the following high-level macro functions:

- (i). Develop and implement institutional strategies, work plans and performance frameworks for data science, advanced analytics, Artificial Intelligence (AI), Machine Learning (ML) and computational research.
- (ii). Provide biostatistical, statistical modelling, computational, bioinformatics and advanced analytical expertise to support research, surveillance and innovation programs.
- (iii). Lead and coordinate the development and application of statistical, predictive, computational and machine learning models for biomedical, clinical, epidemiological and laboratory research.
- (iv). Coordinate analytical data standards, data integration, reproducible analytical workflows and research analytics across relevant institutional programs.
- (v). Develop and support analytical platforms, computational pipelines, dashboards and decision-support tools for evidence generation and institutional decision-making.
- (vi). Provide technical training, mentorship and knowledge transfer in data science, biostatistics, modelling, bioinformatics and advanced analytics in collaboration with the Directorate responsible for Training and Capacity Building.
- (vii). Foster strategic research collaborations and provide technical inputs to resource mobilization for data-intensive and computational research.
- (viii). Provide technical leadership on the analytical aspects of data governance, data quality, data access, model governance and responsible use of advanced analytics, in collaboration with relevant institutional functions.

c) Organizational Structure

The Data Science and Analytics Division will be headed by a Deputy Director, Data Science and Analytics and will comprise three technical sections, each headed by an Assistant Director:

1. *Biostatistics and Statistical Modelling Section*
2. *Bioinformatics and Computational Analytics Section*
3. *Data Analytics and Decision Support Section*

1. Biostatistics and Statistical Modelling Section

a) Rationale

The Section will provide specialized biostatistical, epidemiological and statistical modelling expertise to support the Institute's research, surveillance and innovation programs. It will

contribute to study design, statistical methodology, spatial analysis and modelling of disease patterns, transmission dynamics and intervention outcomes, while providing technical expertise and mentorship for institutional capacity building in collaboration with the Directorate of Capacity Building.

b) Functions

Core mandate: Statistical inference, study design, epidemiology and modelling.

Allocated functions:

- (i). Collect, analyze and interpret biological, clinical, epidemiological and health research data in collaboration with research programs.
- (ii). Contribute to the design, implementation and statistical analysis of pre-clinical, clinical and field epidemiological studies.
- (iii). Provide statistical, methodological and epidemiological consultation and advisory support for research and surveillance activities.
- (iv). Conduct spatial epidemiological analysis and GIS-based assessment of the distribution, ecology and transmission of priority diseases.
- (v). Develop and apply statistical, predictive and machine-learning models to assess disease patterns, intervention outcomes, population impacts and alternative control strategies.
- (vi). Provide specialized technical content, mentorship and training in biostatistics, statistical modelling and epidemiological analysis in collaboration with the Directorate of Capacity Building.

2. Bioinformatics and Computational Analytics Section

a) Rationale

The Section will provide specialized bioinformatics and computational expertise to support laboratory and research programs across the Institute. It will develop and apply computational pipelines, analytical workflows and machine-learning methods for high-dimensional omics, genomic, transcriptomic, proteomic and host–pathogen sequence data. The Section will provide a specialized computational capability for the analysis and interpretation of complex biological data and the development of reproducible computational approaches for biomedical research.

b) Functions

Core mandate: Genomics, omics, bioinformatics and computational biology.

Allocated functions:

- (i). Provide specialized computational pipelines, bioinformatics workflows and analytical support to laboratory and research programs.
- (ii). Develop, manage and maintain computational workflows and analytical frameworks for genomic, transcriptomic, proteomic and other high-dimensional biological datasets.
- (iii). Conduct host–pathogen sequence and genomic analyses to support biomedical and infectious disease research.
- (iv). Develop and implement standardized, reproducible computational workflows and tools for biological and disease research.
- (v). Develop and apply predictive algorithms, Artificial Intelligence (AI) and Machine Learning (ML) methods to identify molecular, genomic and disease patterns.
- (vi). Provide technical support, training and mentorship in bioinformatics, computational analytics and high-throughput genomic data analysis in collaboration with the Directorate of Capacity Building.

3. Data Analytics and Decision Support Section

a) Rationale

The Section will provide expertise in integrated data analytics, data engineering and decision-support systems for research, surveillance and institutional programs. It will develop and maintain analytical data pipelines, data integration and validation workflows, and analytical platforms for transforming diverse scientific data into reliable information for monitoring, evaluation and decision-making. The Section will provide technical support for analytical data governance and responsible use of data and AI/ML systems, in collaboration with relevant institutional functions.

b) Functions

Core mandate: Data integration, data science, advanced analytics and decision support.

Allocated functions:

- (i). Develop and coordinate analytical data standards, data access workflows, data quality procedures and AI/ML model governance in collaboration with ICT, data protection and other relevant institutional functions.
- (ii). Develop and maintain analytical data repositories, metadata frameworks and data integration systems supporting research and institutional analytics.
- (iii). Establish automated data ingestion, cleaning, transformation and validation pipelines for diverse scientific and administrative data sources.

- (iv). Develop data analytics, visualization, dashboards and decision-support tools for research, surveillance, monitoring and evaluation.
- (v). Provide data-driven monitoring and evaluation analytics to assess programme performance, research outputs and institutional activities.
- (vi). Support the dissemination and use of analytical outputs through data portals, reports, scientific publications and other appropriate platforms.

4. Staffing

The Data Science and Analytics Division is staffed as follows in the table below

Designation	Grade KIP	Approved Establishment	In post	Proposed Establishment	Justification
Deputy Director, Data Science & Analytics	3		0	1	To head and lead the division
Chief Research Scientist	2		0	1	To lead resource mobilization and high level modelling.
Senior Principal Research Scientist I	3		0	1	Spearhead advanced modelling and institutional technical mentorship
Sub-total			0	3	Core divisional management team
Biostatistics and Statistical Modelling Section					
Assistant Director	4	1	0	1	To head the section
Senior Principal Research Scientist II	4	2	0	2	Supervise multi-entity statistical reporting and complex study designs.
Principal Biostatistician/Research Scientist	5		0		Retained for career progression
Senior Biostatistician/Research Scientist	6		0		Retained for career progression
Biostatistician/ Research Scientist II/I	8/7	2	2	2	Run data cleaning, GIS scripts and study support.
Sub Total		5	2	5	Manage pre-clinical studies and epidemiological workloads
Bioinformatics and Computational Analytics Section					
Assistant Director	4		0	1	To head the section
Senior Principal Research Scientist II	4		0	1	To design custom computational workflows and advanced laboratory animal models
Principal	5		0		Retained for career

Bioinformatician/ Research Scientist					progression
Senior Bioinformatician/ Research Scientist	6		0	1	To run advanced host pathogen sequence analytics and configure ML/AI models.
Bioinformatician/ Research Scientist II/I	8/7		0	2	To perform raw sequence alignments and map molecular trends for lab investigators.
Sub Total			0	5	To scale up specialized high-throughput processing capabilities for genomic and proteomic platforms.
Data Analytics and Decision Support Section					
Assistant Director	4		0	1	To head the section
Senior Principal Research Scientist II	4		0	2	To implement logical data safeguarding protocols and architect secure automated AI enabled cleaning workflows
Principal Data Scientist/ Research Scientist	5		0		Retained for career progression
Senior Data Scientist/ Research Scientist	6		0		Retained for career progression
Data Scientist/ Research Scientist II/I	8/7		0	2	To administer automated ingestion pipelines and index reference artifacts
Sub Total			0	5	To protect data integrity, automate verification paths, and Secure institutional research memory
Total		5	2	15	To fully operationalize the independent division, correcting critical technical deficits across all three pillars.